

class18

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Table of contents

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2

-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(jsonlite)
```

```
Attaching package: 'jsonlite'
```

```
The following object is masked from 'package:purrr':
```

```
  flatten
```

```
library(lubridate)
```

```
library(datapasta)
```

Q1. With the help of the R “addin” package datapasta assign the CDC pertussis case number data to a data frame called cdc and use ggplot to make a plot of cases numbers over time.

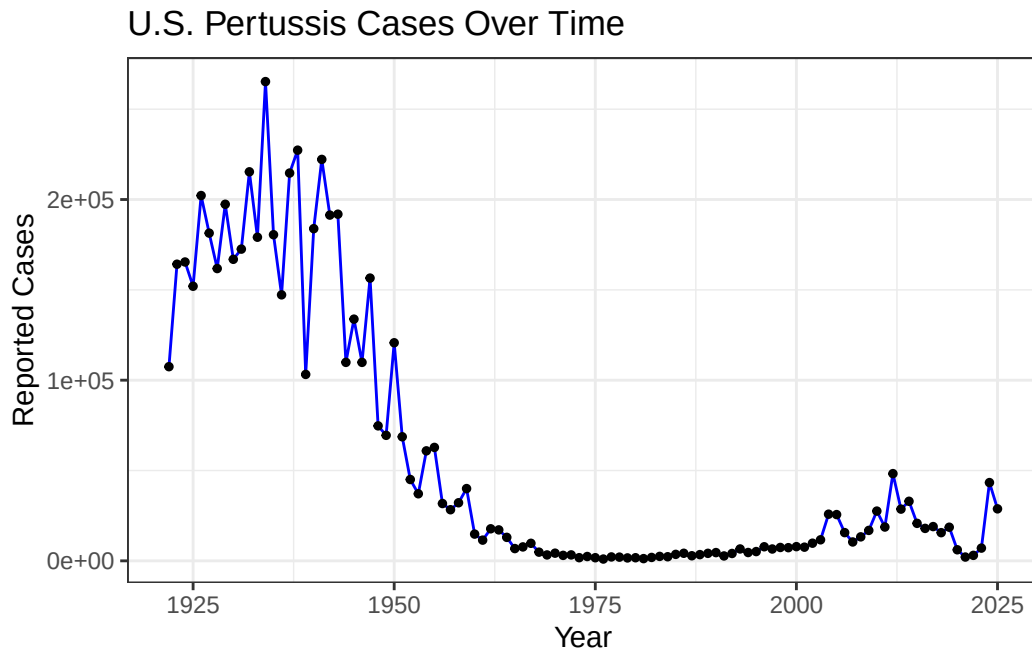
```
# Dataframe was not working properly, so imported data using read.csv
cdc <- read.csv("U.S. Reported Pertussis Cases_ 1922 - 2025.csv")
```

```
# Rename the columns to make them easier to work with
colnames(cdc) <- c("year", "cases", "status")
```

```
cdc$cases <- as.numeric(gsub(",", "", cdc$cases))
cdc$year <- as.numeric(cdc$year)
```

```
# Plot pertussis cases over time
# x-axis = year
# y-axis = number of reported pertussis cases
```

```
ggplot(cdc) +
  aes(x = year, y = cases, group = 1) +
  geom_line(color = "blue") +
  geom_point(size = 1) +
  labs(
    title = "U.S. Pertussis Cases Over Time",
    x = "Year",
    y = "Reported Cases"
  ) +
  theme_bw()
```



Q2. Using the ggplot `geom_vline()` function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine (see example in the hint below). What do you notice?

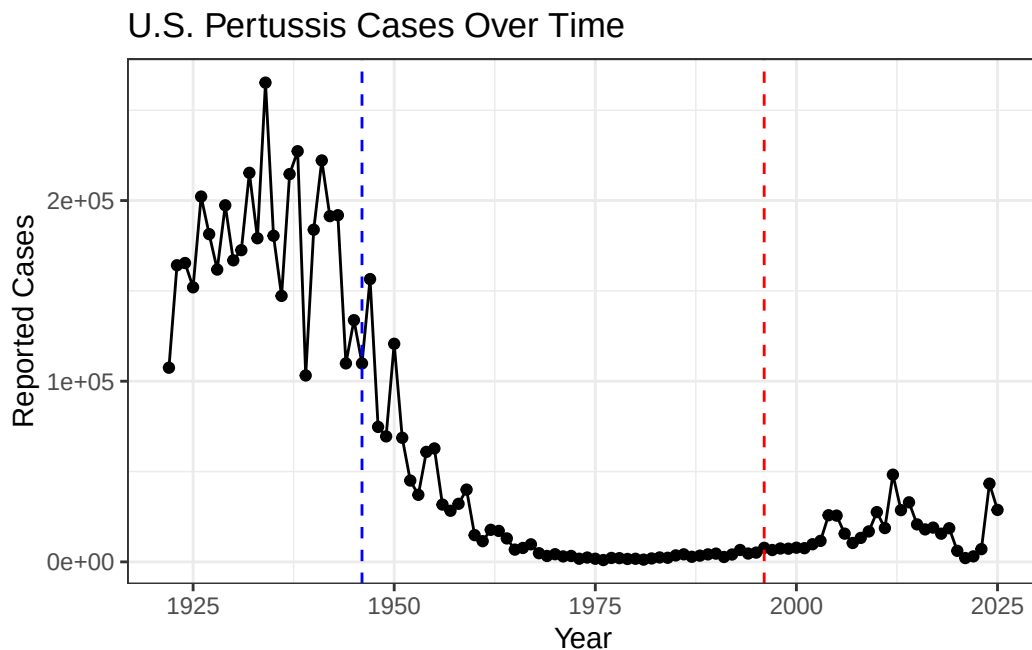
```
ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line() +

  # wP vaccine introduced
  geom_vline(xintercept = 1946,
             color = "blue",
             linetype = "dashed") +

  # switch to aP vaccine
  geom_vline(xintercept = 1996,
             color = "red",
             linetype = "dashed") +

  labs(
    title = "U.S. Pertussis Cases Over Time",
    x = "Year",
    y = "Reported Cases"
```

```
) +  
theme_bw()
```



The command `geom_vline()` draws a vertical line at year 1946, as well as 1996.

Q3. Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend?

After the introduction of the aP vaccine in 1996, pertussis cases began increasing again over time. One possible explanation is that immunity from the aP vaccine wanes faster than immunity from the older wP vaccine. Other possible explanations include increased vaccine hesitancy, improved PCR-based detection methods, and bacterial evolution leading to escape from vaccine-induced immunity.

```
# Load the jsonlite package  
library(jsonlite)
```

```
# Read the subject table from the CMI-PB API  
subject <- read_json(  
  "https://www.cmi-pb.org/api/subject",  
  simplifyVector = TRUE  
)
```

```
# View the first 3 rows of the subject dataframe
head(subject, 3)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	2	wP	Female	Not Hispanic or Latino	White
3	3	wP	Female	Unknown	White

	year_of_birth	date_of_boost	dataset
1	1986-01-01	2016-09-12	2020_dataset
2	1968-01-01	2019-01-28	2020_dataset
3	1983-01-01	2016-10-10	2020_dataset

Q4. How many aP and wP infancy vaccinated subjects are in the dataset?

```
# Count how many subjects received aP vs wP vaccines
table(subject$infancy_vac)
```

```
aP wP
87 85
```

There are 87 who recieved the aP vaccine and 85 who received the wP vaccine.

Q5. How many Male and Female subjects/patients are in the dataset?

```
# Count how many Male and Female subjects are present
table(subject$biological_sex)
```

```
Female    Male
   112     60
```

There are 112 female subjects and 60 male subjects present.

Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc...)?

```
# Create a table showing race breakdown by biological sex
table(subject$biological_sex,
      subject$race)
```

	American Indian/Alaska Native	Asian	Black or African American
Female	0	32	2
Male	1	12	3

	More Than One Race	Native Hawaiian or Other Pacific Islander
Female	15	1
Male	4	1

	Unknown or Not Reported	White
Female	14	48
Male	7	32

Q7. Using this approach determine (i) the average age of wP individuals, (ii) the average age of aP individuals; and (iii) are they significantly different?

```
# Load lubridate package for working with dates
library(lubridate)

# Use today's date and subtract year_of_birth
# to calculate each subject's age in days
subject$age <- today() - ymd(subject$year_of_birth)

# Load dplyr package for filtering and data manipulation
library(dplyr)

# Create a dataframe containing only aP vaccinated individuals
ap <- subject %>%
  filter(infancy_vac == "aP")
```

```
# Display summary statistics for aP ages in years
# round() makes the output cleaner
round(
  summary(
    time_length(ap$age, "years")
  )
)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
23	27	28	28	29	35

```
# Create dataframe containing only wP vaccinated individuals
wp <- subject %>%
  filter(infancy_vac == "wP")
```

```
# Display summary statistics for wP ages in years
round(
  summary(
    time_length(wp$age, "years")
  )
)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
23	33	35	37	40	58

```
# Perform t-test to determine whether
# aP and wP ages are significantly different
x <- t.test(
  time_length(wp$age, "years"),
  time_length(ap$age, "years")
)

# Print p-value
x$p.value
```

```
[1] 2.372101e-23
```

Q8. Determine the age of all individuals at time of boost?

```
# Calculate age at time of booster vaccination by subtracting year_of_birth from date_of_boos
int <- ymd(subject$date_of_boost) -
  ymd(subject$year_of_birth)

# Convert age from days into years
age_at_boost <- time_length(int, "year")

# View first few calculated ages
head(age_at_boost)
```

```
[1] 30.69678 51.07461 33.77413 28.65982 25.65914 28.77481
```

Q9. With the help of a faceted boxplot or histogram (see below), do you think these two groups are significantly different?

```
# Create histogram comparing ages of aP vs wP groups
ggplot(subject) +

  # x-axis = age in years
  # fill color = vaccine type
  aes(time_length(age, "year"),
       fill = as.factor(infancy_vac)) +

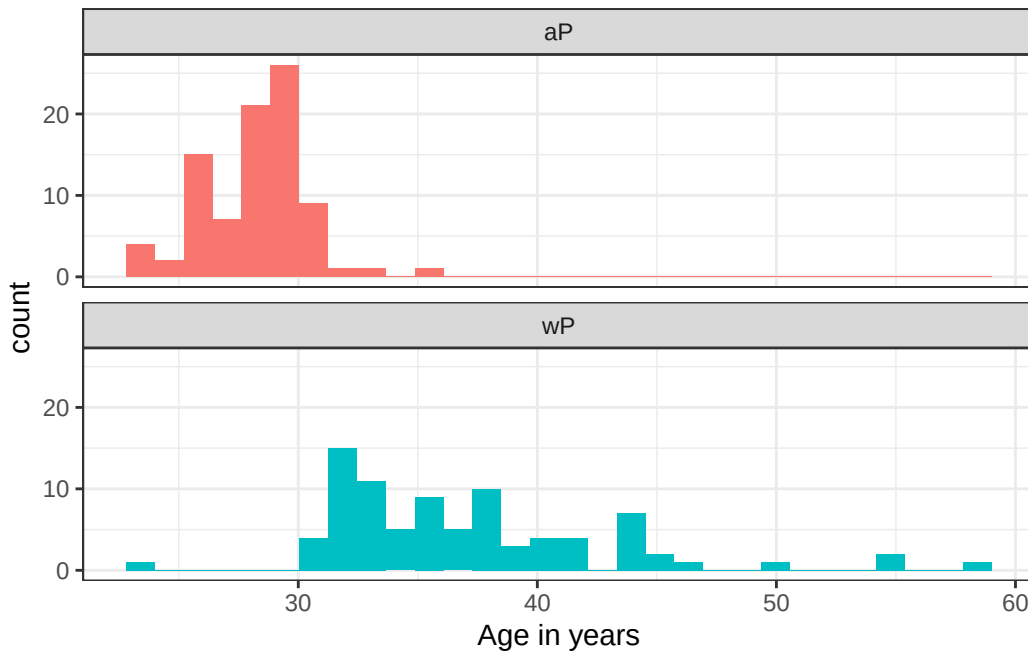
  # Create histogram
  geom_histogram(show.legend = FALSE) +

  # Separate panels for aP and wP groups
  facet_wrap(vars(infancy_vac), nrow = 2) +

  # Label x-axis
  xlab("Age in years") +

  # Use clean theme
  theme_bw()
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



The age distributions between aP and wP groups appear clearly different. The t-test p-value is extremely small, indicating a statistically significant difference between the two groups.

```
# Read the specimen table from the CMI-PB API
# This table contains information about each specimen/sample
specimen <- read_json(
  "https://www.cmi-pb.org/api/specimen",
  simplifyVector = TRUE
)
```

```
# Read the antibody titer table from the current CMI-PB API
# The old endpoint "ab_titer" gives a 404 error, so we use "plasma_ab_titer"
titer <- read_json(
  "https://www.cmi-pb.org/api/plasma_ab_titer",
  simplifyVector = TRUE
)
```

Q9. Complete the code to join specimen and subject tables to make a new merged data frame containing all specimen records along with their associated subject details:

```
# Join specimen with subject information
# inner_join keeps only rows that match in both tables
meta <- inner_join(specimen, subject)
```

Joining with `by = join\_by(subject\_id)`

```
# Check dimensions of merged dataframe
dim(meta)
```

```
[1] 1503  14
```

```
# View first few rows
head(meta)
```

	specimen_id	subject_id	actual_day_relative_to_boost
1	1	1	-3
2	2	1	1
3	3	1	3
4	4	1	7
5	5	1	11
6	6	1	32

	planned_day_relative_to_boost	specimen_type	visit	infancy_vac	biological_sex
1	0	Blood	1	wP	Female
2	1	Blood	2	wP	Female
3	3	Blood	3	wP	Female
4	7	Blood	4	wP	Female
5	14	Blood	5	wP	Female
6	30	Blood	6	wP	Female

	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
4	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
5	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
6	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset

	age
1	14756 days
2	14756 days
3	14756 days
4	14756 days
5	14756 days
6	14756 days

Q10. Now using the same procedure join meta with titer data so we can further analyze this data in terms of time of visit aP/wP, male/female etc.

```
# Join antibody titer data with metadata, this connects antibody measurements to subject info
abdata <- inner_join(titer, meta)
```

Joining with `by = join\_by(specimen\_id)`

```
# Check dimensions of final merged dataframe
dim(abdata)
```

```
[1] 52576    21
```

Q11. How many specimens (i.e. entries in abdata) do we have for each isotype?

```
# Count how many entries there are for each antibody isotype
table(abdata$isotype)
```

```

IgE   IgG  IgG1  IgG2  IgG3  IgG4
6698  5389 10117 10124 10124 10124
```

Q12. What are the different \$dataset values in abdata and what do you notice about the number of rows for the most “recent” dataset?

```
# Count how many rows belong to each dataset
table(abdata$dataset)
```

```

2020_dataset 2021_dataset 2022_dataset 2023_dataset
          31520          8085          7301          5670
```

The dataset column shows which study dataset each row came from. The most recent dataset has fewer rows, likely because newer data collection is less complete or includes fewer samples so far.

Q13. Complete the following code to make a summary boxplot of Ab titer levels (MFI) for all antigens:

```
# Filter the abdata dataframe to include only IgG antibody data
igg <- abdata %>%
  filter(isotype == "IgG")
```

```
# View first few rows of IgG-only dataframe
head(igg)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgG	TRUE	PT	68.56614	3.736992
2	1	IgG	TRUE	PRN	332.12718	2.602350
3	1	IgG	TRUE	FHA	1887.12263	34.050956
4	19	IgG	TRUE	PT	20.11607	1.096366
5	19	IgG	TRUE	PRN	976.67419	7.652635
6	19	IgG	TRUE	FHA	60.76626	1.096457

	unit	lower_limit_of_detection	subject_id	actual_day_relative_to_boost
1	IU/ML	0.530000	1	-3
2	IU/ML	6.205949	1	-3
3	IU/ML	4.679535	1	-3
4	IU/ML	0.530000	3	-3
5	IU/ML	6.205949	3	-3
6	IU/ML	4.679535	3	-3

	planned_day_relative_to_boost	specimen_type	visit	infancy_vac	biological_sex
1	0	Blood	1	wP	Female
2	0	Blood	1	wP	Female
3	0	Blood	1	wP	Female
4	0	Blood	1	wP	Female
5	0	Blood	1	wP	Female
6	0	Blood	1	wP	Female

	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
4	Unknown	White	1983-01-01	2016-10-10	2020_dataset
5	Unknown	White	1983-01-01	2016-10-10	2020_dataset
6	Unknown	White	1983-01-01	2016-10-10	2020_dataset

	age
1	14756 days
2	14756 days
3	14756 days
4	15852 days
5	15852 days
6	15852 days

```
# Create boxplots of normalized antibody titers
# for all antigens across different visits
```

```

ggplot(igg) +

  # x-axis = normalized MFI values
  # y-axis = antigen names
  aes(MFI_normalised, antigen) +

  # Create boxplots
  geom_boxplot() +

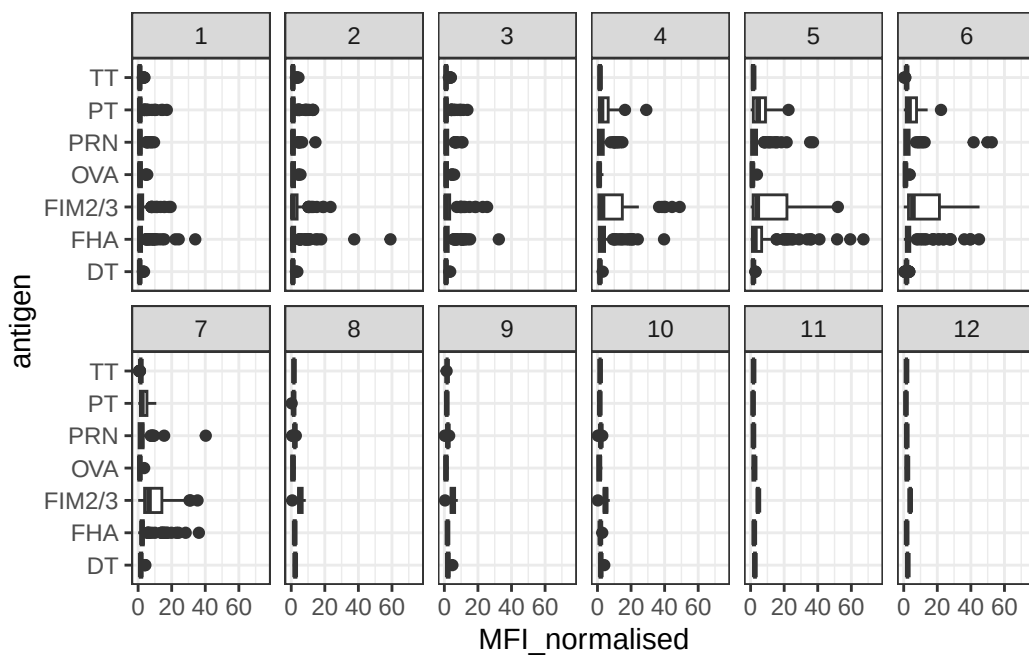
  # Limit x-axis for easier viewing
  xlim(0,75) +

  # Separate plots by visit number
  facet_wrap(vars(visit), nrow = 2) +

  # Use clean plotting theme
  theme_bw()

```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).



Q14. What antigens show differences in the level of IgG antibody titers recognizing them over time? Why these and not others?

Antigens such as PT, FHA, and PRN show clear increases in IgG antibody titers over time after boosting. These antigens are components of pertussis vaccines, so the immune system recognizes and responds strongly to them. Control antigens that are not part of the vaccine show much smaller changes over time.

```
# Create boxplots separated by vaccine type (aP vs wP)
# to compare antibody responses between groups
ggplot(igg) +

  # Color boxplots by infancy vaccine type
  aes(MFI_normalised,
      antigen,
      col = infancy_vac) +

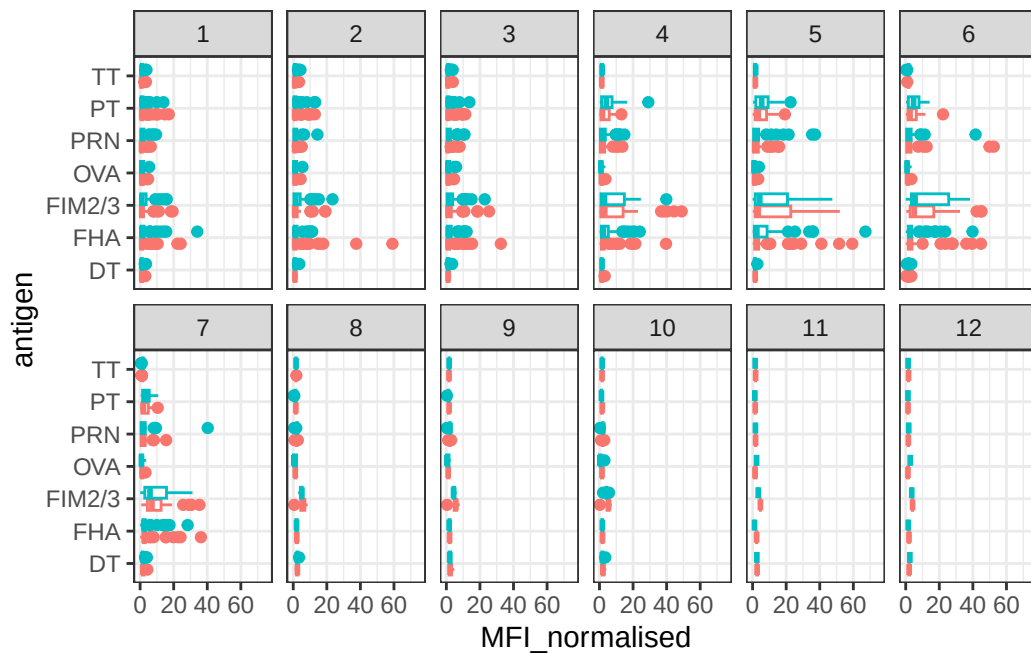
  # Create boxplots
  geom_boxplot(show.legend = FALSE) +

  # Separate by visit number
  facet_wrap(vars(visit), nrow = 2) +

  # Limit x-axis range
  xlim(0,75) +

  # Clean theme
  theme_bw()
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).



```
# Alternative faceted plot
# Separates plots by both vaccine type and visit
igg %>%
  filter(visit != 8) %>%

  ggplot() +

  aes(MFI_normalised,
       antigen,
       col = infancy_vac) +

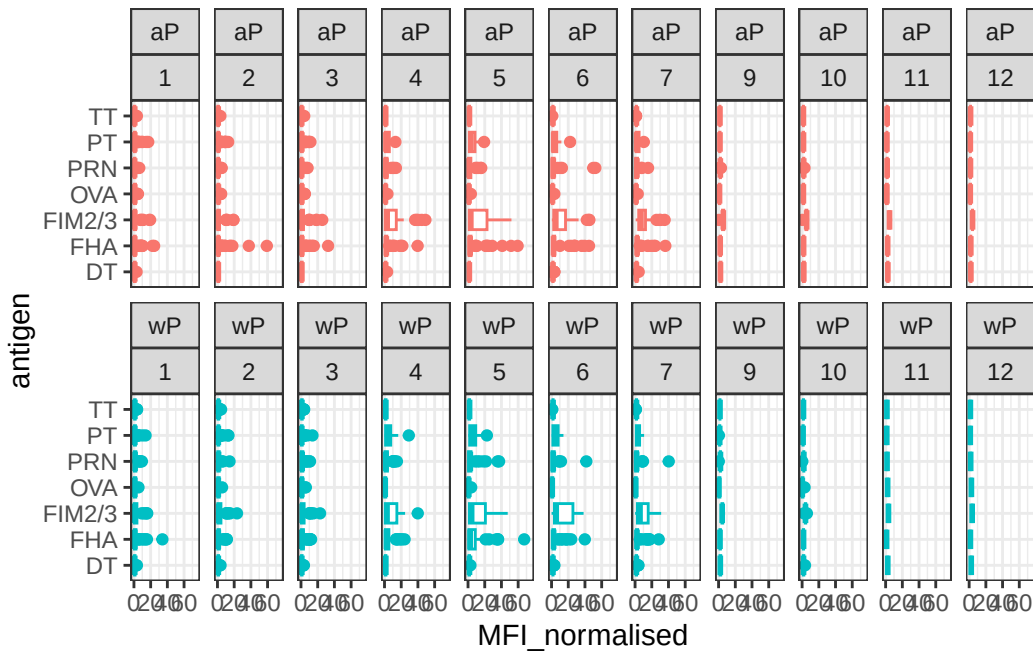
  geom_boxplot(show.legend = FALSE) +

  xlim(0,75) +

  facet_wrap(vars(infancy_vac, visit),
             nrow = 2) +

  theme_bw()
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).



Q15. Filter to pull out only two specific antigens for analysis and create a boxplot for each. You can chose any you like. Below I picked a “control” antigen (“OVA”, that is not in our vaccines) and a clear antigen of interest (“PT”, Pertussis Toxin, one of the key virulence factors produced by the bacterium *B. pertussis*).

```
# Plot OVA antigen responses
# OVA acts as a control antigen
filter(igg, antigen == "OVA") %>%

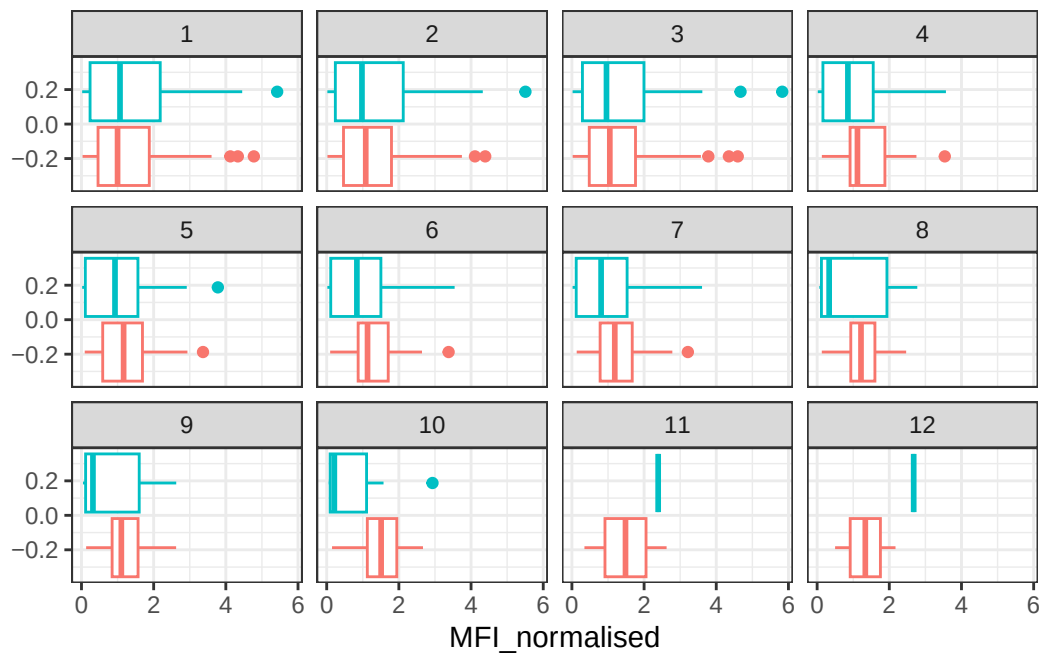
ggplot() +

aes(MFI_normalised,
     col = infancy_vac) +

geom_boxplot(show.legend = FALSE) +

facet_wrap(vars(visit)) +

theme_bw()
```

```
# Plot FIM2/3 antigen responses
filter(igg, antigen == "FIM2/3") %>%

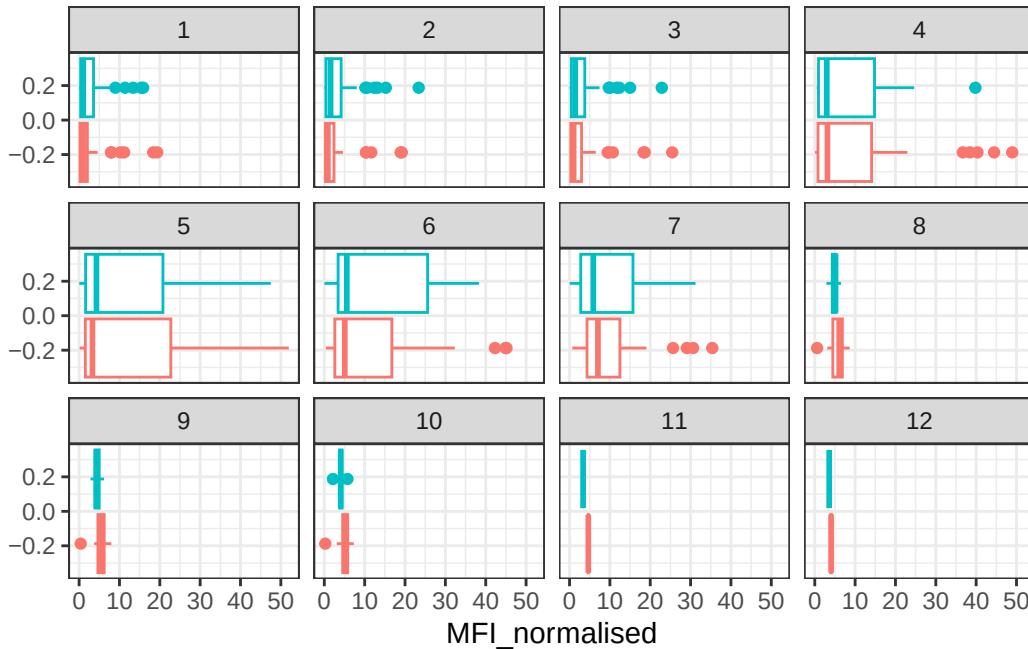
ggplot() +

aes(MFI_normalised,
     col = infancy_vac) +

geom_boxplot(show.legend = FALSE) +

facet_wrap(vars(visit)) +

theme_bw()
```



Q16. What do you notice about these two antigens time courses and the PT data in particular?

PT and FIM2/3 antibody levels increase after boosting, peak around visit 5, and then decline over time. OVA levels remain relatively low because OVA is not part of the pertussis vaccine and acts as a control antigen.

Q17. Do you see any clear difference in aP vs. wP responses?

There do not appear to be dramatic differences between aP and wP responses overall, although some antigens may show subtle differences in antibody magnitude or variability.

```
# Filter for only the 2021 dataset
abdata.21 <- abdata %>%
  filter(dataset == "2021_dataset")
```

```
# Plot PT IgG antibody levels over time for 2021 dataset
abdata.21 %>%

  filter(isotype == "IgG",
         antigen == "PT") %>%

  ggplot() +
```

```

aes(
  x = planned_day_relative_to_boost,
  y = MFI_normalised,
  col = infancy_vac,
  group = subject_id
) +

# Add points for each measurement
geom_point() +

# Connect points for each subject
geom_line() +

# Add dashed vertical line for day 0
geom_vline(xintercept = 0,
           linetype = "dashed") +

# Add dashed vertical line for day 14
geom_vline(xintercept = 14,
           linetype = "dashed") +

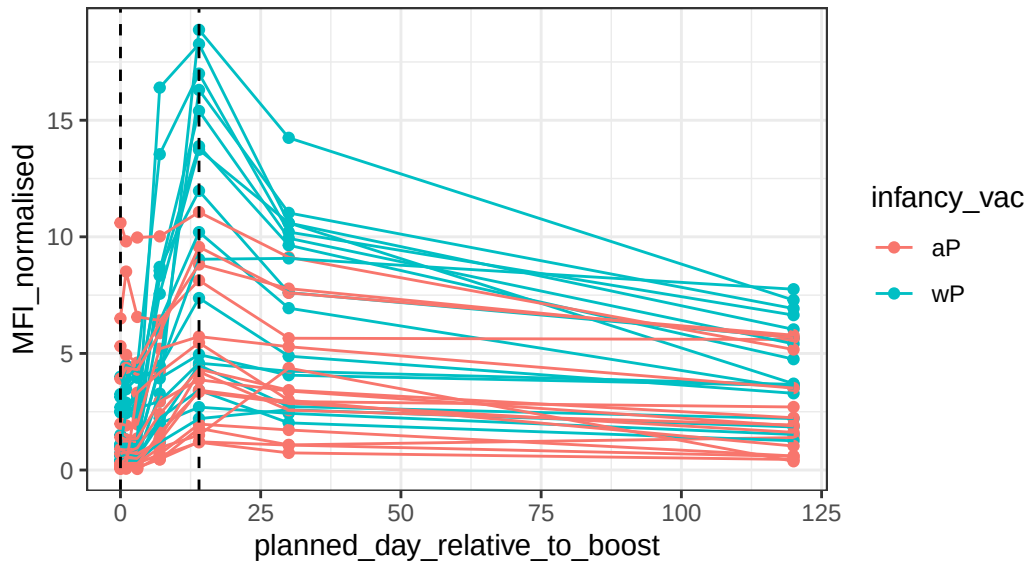
labs(
  title = "2021 dataset IgG PT",
  subtitle = "Dashed lines indicate day 0 and day 14"
) +

theme_bw()

```

2021 dataset IgG PT

Dashed lines indicate day 0 and day 14



Q18. Does this trend look similar for the 2020 dataset?

The 2020 dataset shows a similar overall trend, where PT antibody levels increase after boosting, peak around two weeks, and then gradually decline.

Q19. Make a plot of the time course of gene expression for IGHG1 gene (i.e. a plot of visit vs. tpm).

```
# Create URL for RNA-seq API query
# This specific gene ID corresponds to IGHG1
url <- "https://www.cmi-pb.org/api/v2/rnaseq?versioned_ensembl_gene_id=eq.ENS00000211896.7"

# Read RNA-seq data from the API
# simplifyVector = TRUE converts JSON into dataframe format
rna <- read_json(
  url,
  simplifyVector = TRUE
)

# Join RNA-seq data with metadata table
# This connects gene expression values with subject information
ssrna <- inner_join(rna, meta)
```

Joining with `by = join\_by(specimen\_id)`

```
# View first few rows of merged RNA dataset
head(ssrna)
```

	versioned_ensembl_gene_id	specimen_id	raw_count	tpm	subject_id
1	ENSG00000211896.7	344	18613	929.640	44
2	ENSG00000211896.7	243	2011	112.584	31
3	ENSG00000211896.7	261	2161	124.759	33
4	ENSG00000211896.7	282	2428	138.292	36
5	ENSG00000211896.7	345	51963	2946.136	44
6	ENSG00000211896.7	244	49652	2356.749	31

	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type
1	3	3	Blood
2	3	3	Blood
3	92	120	Blood
4	1	1	Blood
5	7	7	Blood
6	7	7	Blood

	visit	infancy_vac	biological_sex	ethnicity	race
1	3	aP	Female	Hispanic or Latino	More Than One Race
2	3	wP	Female	Not Hispanic or Latino	Asian
3	7	wP	Male	Hispanic or Latino	More Than One Race
4	2	aP	Female	Hispanic or Latino	White
5	4	aP	Female	Hispanic or Latino	More Than One Race
6	4	wP	Female	Not Hispanic or Latino	Asian

	year_of_birth	date_of_boost	dataset	age
1	1998-01-01	2016-11-07	2020_dataset	10373 days
2	1989-01-01	2016-09-26	2020_dataset	13660 days
3	1990-01-01	2016-10-10	2020_dataset	13295 days
4	1997-01-01	2016-10-24	2020_dataset	10738 days
5	1998-01-01	2016-11-07	2020_dataset	10373 days
6	1989-01-01	2016-09-26	2020_dataset	13660 days

```
# Plot time course of IGHG1 gene expression
```

```
ggplot(ssrna) +
```

```
# x-axis = visit number
```

```
# y-axis = TPM gene expression
```

```
# group by subject so each person's measurements connect
```

```
aes(visit,
```

```
    tpm,
```

```

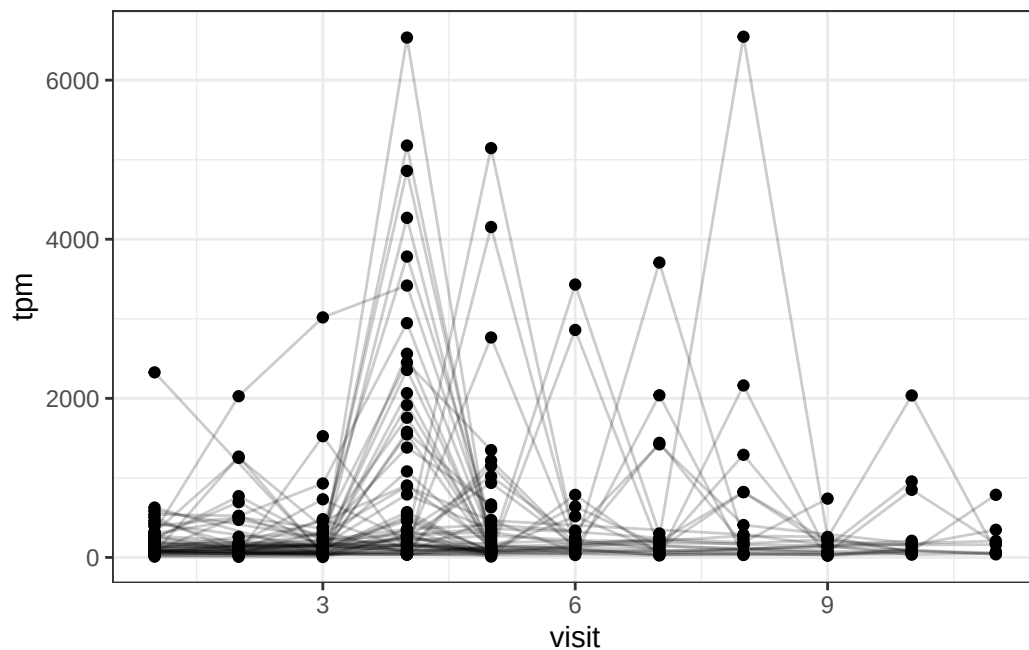
group = subject_id) +

# Add points for each observation
geom_point() +

# Connect points with lines
geom_line(alpha = 0.2) +

# Clean plotting theme
theme_bw()

```



Q20.: What do you notice about the expression of this gene (i.e. when is it at it's maximum level)?

IGHG1 gene expression appears to increase after boosting and reaches its highest levels around the middle visits before decreasing again at later visits.

Q21. Does this pattern in time match the trend of antibody titer data? If not, why not?

The RNA expression pattern does not perfectly match antibody titer levels because RNA expression occurs earlier than production and accumulation of antibodies. Antibodies are proteins that can remain in circulation for longer periods after gene expression decreases.

```

# Create boxplots comparing RNA expression
# between aP and wP groups across visits
ggplot(ssrna) +

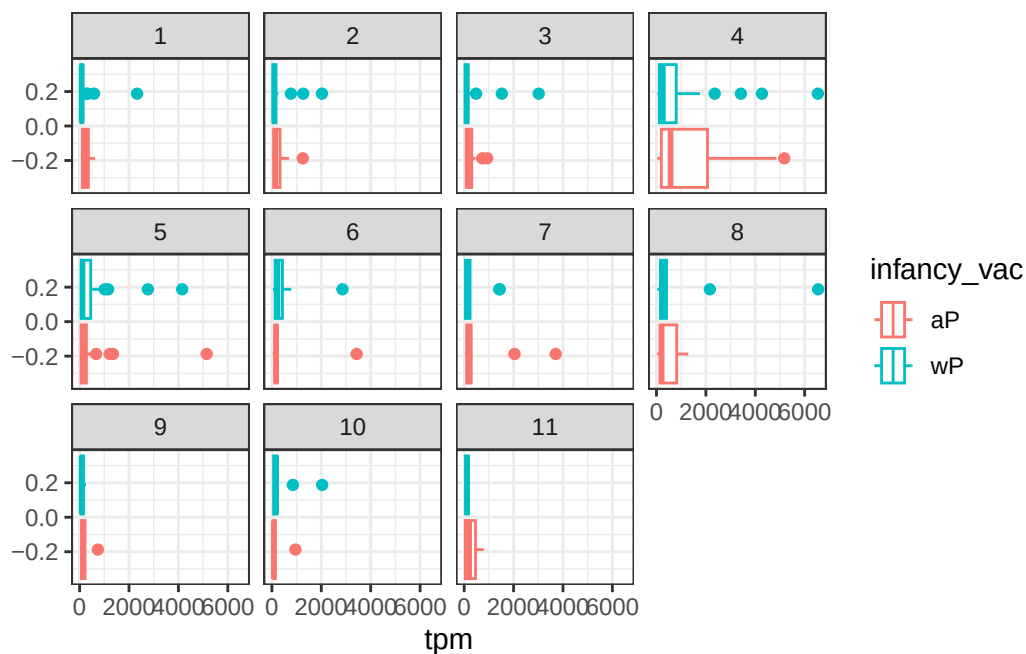
  # x-axis = TPM expression
  # color = vaccine type
  aes(tpm,
      col = infancy_vac) +

  # Create boxplots
  geom_boxplot() +

  # Separate plots by visit
  facet_wrap(vars(visit)) +

  # Use clean theme
  theme_bw()

```



```

# Focus only on visit 4
# Compare TPM density distributions between vaccine groups
ssrna %>%

```

```

filter(visit == 4) %>%

ggplot() +

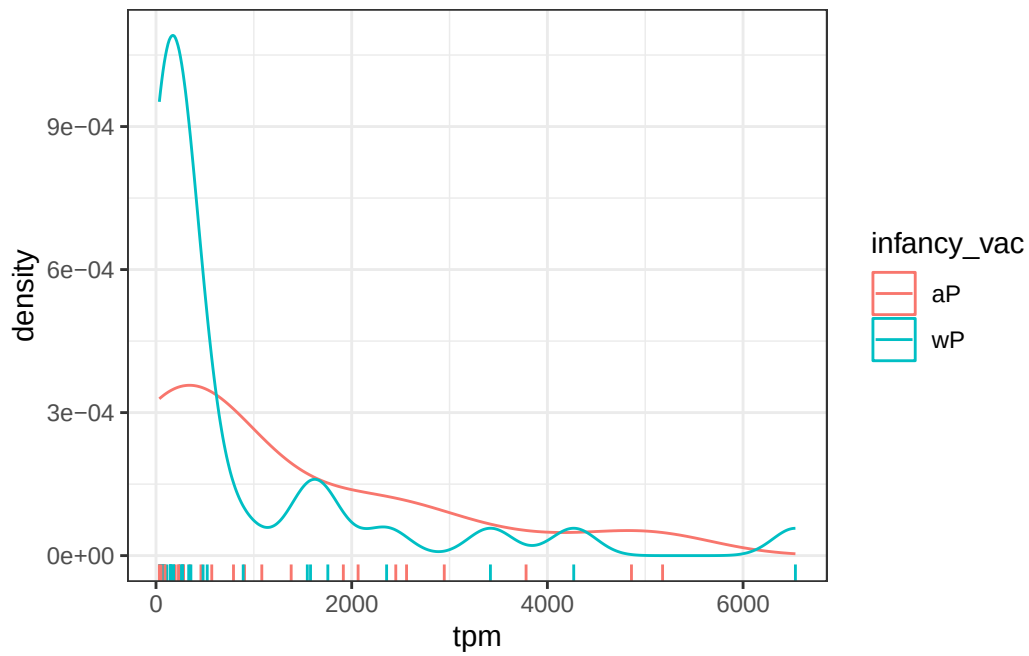
aes(tpm,
     col = infancy_vac) +

# Density curves
geom_density() +

# Add rug plot showing individual observations
geom_rug() +

# Clean theme
theme_bw()

```



There does not appear to be a major difference in IGHG1 expression between aP and wP groups, even when focusing on a specific visit.